



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
10-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 222

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A cubical box is used to keep cold drinks in a bus. Area of each face of cubical box is 0.80 m^2 and each wall has thickness 2.00 cm . It is filled with ice and water which remains at 0°C . Temperature outside each wall is 30°C then: (Thermal conductivity of material of box = $1 \times 10^{-2} \text{ W/m-k}$) (Latent heat of fusion of ice = $3.2 \times 10^5 \text{ J/kg}$)

(A) Total heat current in the box is 60 J/s

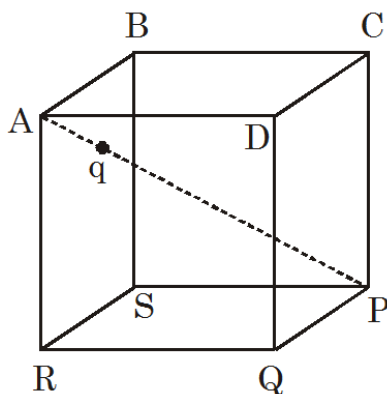
(B) Total heat current in the box is 72 J/s

(C) Mass of ice melted in 1 hour is 810 gm

(D) Mass of ice melted in 1 hour is 690 gm

Space for Rough Work

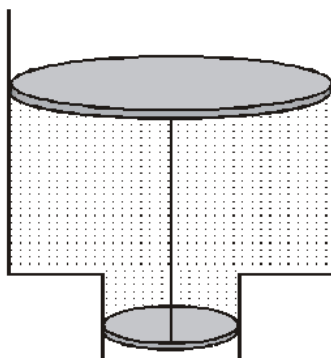
2. A charge 'q' is placed on the diagonal AP of a cube at a distance $\frac{AP}{3}$ from the point A. Choose the correct options.



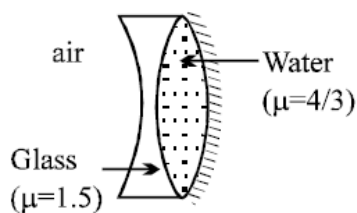
- (A) the sum of electric flux passing through the surfaces ABCD and PQRS is $\frac{q}{3\epsilon_0}$
- (B) the sum of electric flux passing through the surfaces ABCD and PQRS is $\frac{q}{8\epsilon_0}$
- (C) the flux through both the surfaces ABCD and PQRS are same.
- (D) the flux through the surfaces ABCD is larger than the flux through surface PQRS.
3. A sound wave of frequency n travels horizontally to the right. It is reflected from a large vertical plane surface moving to the left with a speed v . The speed of sound in the medium is c . Then
- (A) The number of waves striking the surface per second is $\frac{n(c+v)}{c}$
- (B) The frequency of the reflected wave is $\frac{n(c+v)}{(c-v)}$
- (C) The wavelength of the reflected wave is $\frac{c(c-v)}{n(c+v)}$
- (D) The number of beats heard by a stationary listener to the left of the reflecting surface is $\frac{nv}{c-v}$

Space for Rough Work

4. A vertical cylinder contains ideal gas with the help of two pistons whose combined mass is M . Pistons are connected by an inextensible string. The difference in the area of pistons is A . Atmospheric pressure is P_0 . Initially the system is in equilibrium as shown in the figure. Now heat is supplied to the system. Mark the correct statement(s)



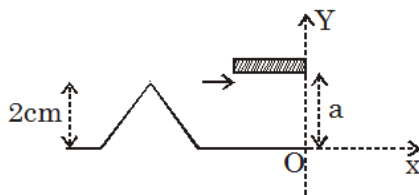
- (A) Volume will increase
(B) Pressure will remain constant
(C) Temperature will increase
(D) Internal energy of gas will remain constant.
5. The radius of curvature of the left and right surface of the concave lens are 10 cm and 15 cm respectively. The radius of curvature of the mirror is 15 cm



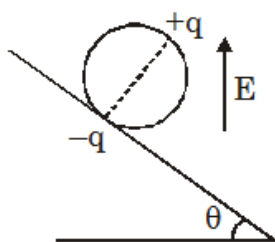
- (A) Equivalent focal length of the combination is -18cm
(B) Equivalent focal length of the combination is $+36\text{ cm}$
(C) The system behaves like a concave mirror
(D) The system behaves like a convex mirror

Space for Rough Work

6. In the figure, a wave pulse is moving on a string towards a free end and there is no loss of energy on reflection. A touch screen cell phone is kept parallel to x-axis. Cell phone is at a distance 'a' from the origin. For what value of 'a' (in cm) will the cell phone screen get activated.



- (A) 3 (B) 5.5 (C) 3.5 (D) 2
7. Two point charges $+q$ and $-q$ are fixed on diametrically opposite point on the uniform ring, such that $-q$ is at bottom and is in contact with a perfect insulator incline plane. Total mass is equal to m . It remains in equilibrium on the rough inclined plane in the presence of uniform vertical electric field $\vec{E}$. Then :-



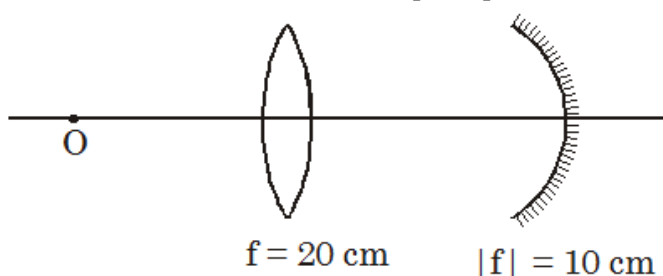
- (A) The value of friction force is $mg\sin\theta$
- (B) The value of friction force is less than $mg\sin\theta$
- (C) The value of electric field is $\frac{mg}{2q}$
- (D) The value of electric field is $\frac{mg \tan \theta}{2q}$

Space for Rough Work

8. Choose the correct statement(s) :-

- (A) The image of sun can not be produced at center of glass sphere, (assuming the space between sphere and sun is filled with air).
- (B) If a lens having $\mu = 1.5$ has focal length 20 cm in air, then its focal length will be 60 cm when it is immersed in water of refractive index $\frac{4}{3}$
- (C) The laws of reflection and refraction are true for all surface and pairs of media at the point of incidence.
- (D) The thick lenses give coloured images due to dispersion of light.

9. Convex lens and a concave mirror having same axis are kept at a distance of 40 cm from each other. A point object is kept at 20 cm from the lens on the common principal axis.



- (A) The image is formed to the left of the object
 - (B) If lens is shifted to the left, the image can be formed at the object.
 - (C) If the mirror is shifted to the left, the image can be formed at the object.
 - (D) If the object is shifted to the left, the image can be formed at the object.
10. A small light source moves at a constant speed v along a straight line making an angle 10° with the main optical axis of the converging lens. Image of the source also moves along straight line, which is at an angle 20° with the main optical axis of the lens. Find the minimum image speed (u_{rel}) relative to the source. What's the magnification at this moment ?

(A) $u_{\text{rel}} = \frac{v}{2}$

(B) $u_{\text{rel}} = v \frac{\sqrt{3}}{2}$

(C) $m = -0.5$

(D) Image must be real

Space for Rough Work

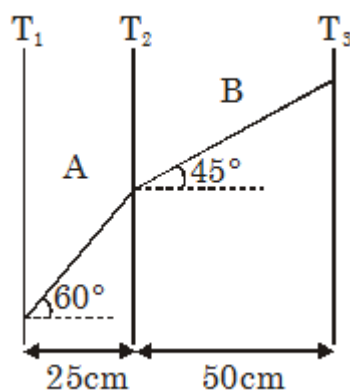
SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

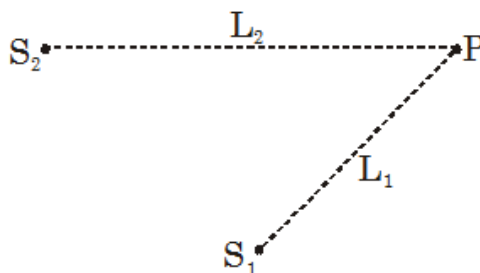
1. The volume of glass vessel is 1000 cc at 20°C . What volume of mercury should be poured into it at this temperature so that the volume of the remaining space does not change with temperature is $10n$ cc. Coefficient of cubical expansion of mercury and glass are $1.8 \times 10^{-4} / ^\circ\text{C}$ and $9.0 \times 10^{-6} / ^\circ$ respectively. Find the value of n .
2. Two conductors A and B each of cross section area 5 cm^2 are connected in series. Variation of temperature (in $^\circ\text{C}$) along the length (in cm) is as shown in the figure. If thermal conductivity of A is $200 \text{ J/m-sec-}^\circ\text{C}$ and the heat current in the conductors is 0.173 J/s . The thermal conductivity of B is $K \times 50 \sqrt{3} \text{ J/m-sec}^\circ\text{C}$. Find the value of K .



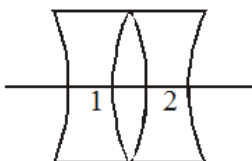
3. The minimum deviation suffered by a ray emerging through a prism $\left(\mu = \frac{2}{\sqrt{3}}\right)$ is equal to half the refracting angle of the prism. Find the value of C if the refracting angle of the prism is $(2\pi/C)$.

Space for Rough Work

4. Two sound sources shown in the figure vibrate in phase. By moving S_1 along PS_1 consecutive minima are heard when $L_1 - L_2$ has values, 20 cm, 60 cm and 100 cm. If the frequency of sound source is $\frac{1700}{n}$ Hz. Then find the value of n ; [Speed of sound is 340 m/s]



5. Two identical metallic sheets each behaves like a black body are kept parallel to each other with small separation between them in vacuum. Thermal energy is generated at a constant rate P in one of the sheets by a passing electricity in it. In steady state, temperature of the other sheet is found to be constant at 300 K. Find the value of P in kilo-Watt. Both surfaces of first sheet are at same temperature. (Given area of the plates $A = \frac{10^3}{17 \times 27} \text{ m}^2$ and $\sigma = \frac{17}{3} \times 10^{-8} \text{ W/m}^2 \text{ K}^4$)
6. Two identical equi-concave lenses made of glass of refractive index 1.5, placed in contact has power P . When a liquid of refractive index μ is filled in the gap between the concave lenses, the power becomes of initial value ($p/3$). The value of μ is $K/3$. Find the value of K .



Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

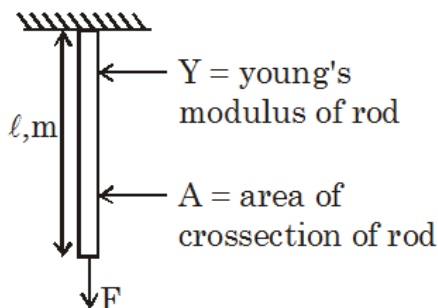
- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 In all other cases

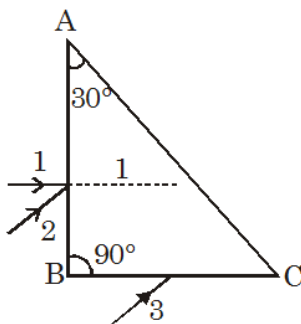
1. Match the following



Column-I		Column-II	
(A)	Stress at a distance x from lower end	(P)	$\frac{1}{2} \frac{Y A}{l} (\Delta l)^2$
(B)	Speed of longitudinal wave in rod	(Q)	$\sqrt{\frac{\ell F + mgx}{m}}$
(C)	Speed of transverse wave in rod at a distance x from lower end	(R)	$\frac{F + \frac{mgx}{l}}{A}$
(D)	If elongation of rod is Δl then elastic pot. energy in the rod will be	(S)	$\sqrt{\frac{Y A \ell}{m}}$
		(T)	None of these

Space for Rough Work

2. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC. Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then match the following.



Column-I		Column-II	
(A)	Refractive index of the material of prism is	(P)	1.20
(B)	Angle of incidence in degree of ray (2) is	(Q)	90
(C)	Deviation in degree suffered by ray (2) is	(R)	2
(D)	Deviation in degree suffered by ray (3) is	(S)	60
		(T)	1.5

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

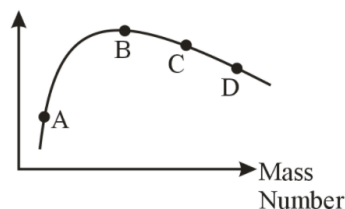
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. 0.1 mol H_3A is added to 10 litre of buffer solution maintained at $pH = 2$. Select the correct statement(s)
[Given : $K_{a1} = 10^{-4}$, $K_{a2} = 10^{-7}$, $K_{a3} = 10^{-11}$ for H_3A]

- (A) % ionisation of H_3A is almost 10% (B) % ionisation of H_3A is almost 1%
(C) $[A^{3-}]$ is almost 10^{-13} % of initial H_3A (D) $[A^{3-}]$ is almost 10^{-14} % of initial H_3A

2. BE per Nucleon



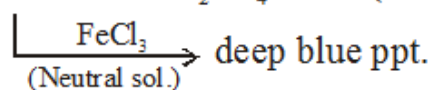
Which of the following nuclear reaction must be exothermic

- (A) $2C \rightarrow B + D$ (B) $D \rightarrow B + C$ (C) $2A \rightarrow B$ (D) $B \rightarrow A + C$

Space for Rough Work

3. During electrolysis mass of a metal produced is proportional to -
- (A) Time duration of current and current efficiency
 (B) Strength of current and molar mass of substance
 (C) Anode surface area and its porosity
 (D) Initial oxidation state of metal and its molar mass
4. Which of the following is /are reversible cell
- (A) $\text{Sb} | \text{Sb}_2\text{O}_3 | \text{OH}^- || \text{Fe}^{2+}, \text{Fe}^{3+} | \text{Pt}$ (B) $\text{Ag} | \text{AgCl} | \text{Cl}^- || \text{Ag}^+ | \text{Ag}$
 (C) $\text{Pt} | \text{Hg} | \text{Hg}_2\text{Cl}_2 | \text{Cl}^- | \text{Cl}_2 | \text{Pt}$ (D) $\text{Zn} | \text{H}_2\text{SO}_4 | \text{Cu}$

5. Salt 'X' + conc. $\text{H}_2\text{SO}_4 \rightarrow \text{Y}$ (Heterodiatomic gas)



Salt 'X' + CuSO_4 solution $\rightarrow$ chocolate brown ppt.

Based on above observations select the **CORRECT** option :-

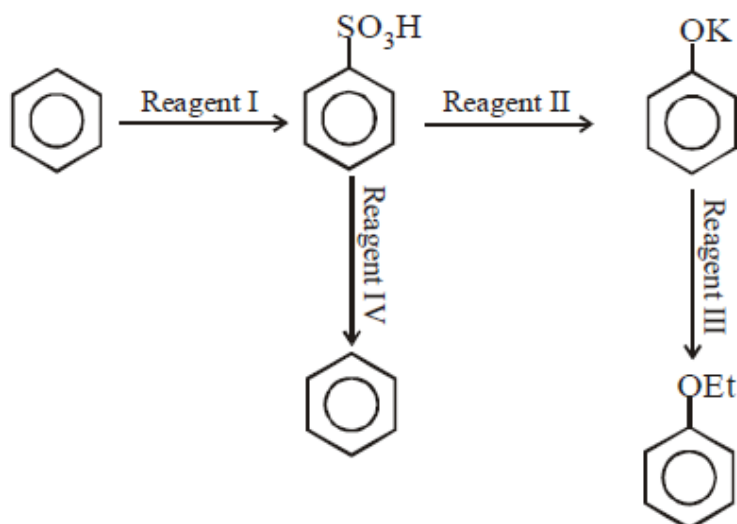
- (A) Central atom in anion of salt 'X' is sp^3d^2 hybridised
 (B) Central atom in anion of salt 'X' belongs to '3d' series
 (C) Salt 'X' gives HCN gas with dil. H_2SO_4
 (D) Gas 'Y' is acidic in nature

Space for Rough Work

6. Which of the following statement(s) is/are **CORRECT**?
- (A) In Ostwald process for the preparation of HNO_3 , Pt/Rh is used as catalyst.
- (B) In Contact process for the preparation of H_2SO_4 , V_2O_5 is used as catalyst.
- (C) In $[\text{Cu}(\text{CN})_4]^{3-}$, value of EAN is 36.
- (D) In alkaline medium, $\text{H}_2\text{S}_{(\text{g})}$ gives positive test with sodium nitroprusside solution.
7. Which of the properties are **INCORRECT** regarding NaHCO_3 and KHCO_3 .
- (A) Both are equally soluble in water
- (B) Both have similar structure of HCO_3^- ion in their solid
- (C) Both solid consists of ion-dipole type of interaction
- (D) Both on heating produces respective carbonate as residue
8. Choose the incorrect option ?
- (A) Polythene is a chain growth polymer
- (B) Bithional is added to soaps to impart antiseptic properties
- (C) Cimetidine is used as antacid
- (D) Penicilline(G) is an example of limited spectrum antibiotic

Space for Rough Work

9. Correct statement(s) regarding following sequence reaction is/are :



- (A) Reagent I is $\text{HNO}_3 + \text{H}_2\text{SO}_4$ (B) Reagent II is fused KOH
 (C) Reagent III is Et_2SO_4 (D) Reagent IV is H_3O^+ (dilute acid)
10. Which of the following compounds reduces Tollen's reagent?
 (A) Glucose (B) Sucrose (C) Fructose (D) Galactose

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. Calculate the pressure exerted (in atm) by 1 mole of CH_4 gas at a temperature of $\frac{24}{0.0821} \text{ K}$ if volume occupied by CH_4 molecules is negligible. Given $a = 2 \text{ atm lit}^2 \text{ mol}^{-2}$.
[Give your answer as sum of digits and Fill OMR sheet]

2. Find (O–O) bond enthalpy in H_2O_2 (kJ/mol)

Given : $\Delta H_f(\text{H}_2\text{O}_2, l) = -200 \text{ kJ/mol}$

$\Delta H_f(\text{H}_2\text{O}, l) = -285 \text{ kJ/mol}$

$\Delta H_{\text{vap.}}(\text{H}_2\text{O}_2, l) = 60 \text{ kJ/mol}$

$\Delta H_{\text{vap.}}(\text{H}_2\text{O}, l) = 40 \text{ kJ/mol}$

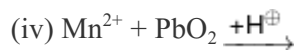
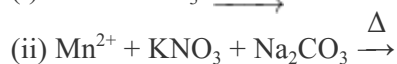
$\Delta H_{\text{atomisation}}(\text{O}_2, g) = 300 \text{ kJ/mol}$

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

3. Total number of reagent/s that can convert I^- into I_2
 Cu^{+2} , Fe^{+3} , Hg^{+2} , NO_2^- , Cl_2 , Br_2 , Fe^{+2}

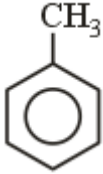
Space for Rough Work

4. Find out total reaction(s) with their given condition(s), which converts Mn^{2+} into MnO_4^- .



5. Glucose $\xrightarrow[\text{(iv) NO}_2\text{BF}_4]{\text{(i) P+HI} / \Delta}$ 'X'
 (Major)
 (ii) $\text{Al}_2\text{O}_3 + \text{V}_2\text{O}_5 / \Delta$
 (iii) $\text{CO} + \text{HCl} / \text{AlCl}_3$

What is the position of prefix in formed major product (X)?

6.  $\xrightarrow[\text{(ii) H}^+]{\text{(i) KMnO}_4 / \text{OH}^-}$ X $\xrightarrow[\text{(ii) Electrolysis}]{\text{(i) KOH}}$ Y (Aromatic hydrocarbon)

In 'Y' total number of sigma bond formed by overlapping of $(\text{sp}^2 - \text{sp}^2)$ orbital(s) = (P)

Total number of πe^- in 'Y' = (Q)

value of $|P - Q|$ is

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 In all other cases

1.

Column-I		Column-II	
(A)	$\text{S}_2\text{O}_3^{2-}$	(P)	give fumes/gas with dil H_2SO_4
(B)	NO_3^-	(Q)	give violet coloured ppt. with $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$
(C)	NO_2^-	(R)	give white ppt with BaCl_2 solution
(D)	S^{2-}	(S)	identified by brown ring test
		(T)	identified by methylene blue dye test

Space for Rough Work

2.

Column - I (Compound)		Column - II (Can be distinguish by)	
(A)	CH ₃ -CHO & Ph-CHO	(P)	Br ₂ / H ₂ O test
(B)	CH ₃ -CH ₃ & HC ≡ CH	(Q)	Ammonical AgNO ₃
(C)	CH ₃ -C≡CH & $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3-\text{C}-\text{CH}_3 \end{array}$	(R)	Iodoform test
(D)	CH ₃ -CHO & $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3-\text{C}-\text{CH}_3 \end{array}$	(S)	NaHSO ₃ test
		(T)	Fehling test

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $1, z_1, z_2, \dots, z_{10}$ are the roots of equation $z^{11} - 1 = 0$, then-
(where $i = \sqrt{-1}$ and $\omega (\neq 1)$ is complex cube root of unity)

(A) $\prod_{k=1}^{10} (i - z_k) = i$ (B) $\prod_{k=1}^{10} (i - z_k) = -i$ (C) $\prod_{k=1}^{10} (\omega + z_k) = \omega$ (D) $\prod_{k=1}^{10} (\omega + z_k) = \omega^2$

2. Let P be a 2×2 matrix such that $P \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $P^2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The value(s) of p for which $\det(P - pI) = 0$, where I is an identity matrix of order 2, is/are
[Note : $\det(M)$ denotes determinant of square matrix M]

(A) 0 (B) -1 (C) 1 (D) 2

Space for Rough Work

3. Let A be a fourth order diagonal matrix whose entries are complex numbers such that $A^4 = I_4$. If trace of A is zero then (here I_4 is fourth order unit matrix)
- (A) There exist 24 such matrices A (B) There exist 36 such matrices A
 (C) Determinant value of all such matrices is 1 (D) Determinant value of all such matrices is -1
4. A, B & C be three points in space such that $A \equiv (3, 4, 2)$, $B \equiv (3, 0, 2)$ & $C \equiv (5, 0, 0)$, then :-
- (A) distance of orthocentre of ΔABC from origin is $\sqrt{13}$
 (B) the position vector of circumcentre of ΔABC is $4\hat{i} + 2\hat{j} + \hat{k}$
 (C) if point D forms parallelogram with remaining three points, then $D \equiv (5, 4, 0)$
 (D) if point D forms parallelogram with remaining three points then $D \equiv (5, -4, 0)$
5. If the complex numbers z_1, z_2, z_3 and z_4 , in order, are vertices of a quadrilateral such that $5z_1 - 6z_2 + 3z_3 - 2z_4 = 0$ and $5|z_1 - z_4|^2 = 9|z_2 - z_3|^2$ then which of the following is/are correct.
- (A) z_1, z_2, z_3 and z_4 are vertices of a rectangle.
 (B) The line segment joining z_1 and z_3 divides the line segment joining z_2 and z_4 in the ratio 1 : 3.
 (C) The line segment joining z_2 and z_4 divides the line segment joining z_1 and z_3 in the ratio 3:5.
 (D) z_1, z_2, z_3 and z_4 are concyclic.
6. If a, b, c are distinct integers and $\omega \neq 1$ is a cube root of unity, then minimum value of " $|a + b\omega + c\omega^2| + |a + b\omega^2 + c\omega|$ " is greater than
- (A) $\sqrt{3}$ (B) $2\sqrt{2}$ (C) $\sqrt{2}$ (D) $\sqrt{6}$

Space for Rough Work

7. Three different numbers are chosen at random without replacement from $\{1, 2, 3, \dots, 15\}$, Let E_1 be the event that minimum of the chosen numbers is 5 and E_2 be the event that their maximum is 10, then
- (A) $P(E_1) = \frac{9}{91}$ (B) $P(E_2) = \frac{36}{455}$
- (C) $P(E_1 \cap E_2) = \frac{4}{455}$ (D) $P\left(\frac{E_1}{E_2}\right) = \frac{1}{9}$
8. Let X be a set containing 'n' elements. If two subsets A and B of X are picked at random, the probability that A and B have the same number of elements is
- (A) $\frac{{}^{2n}C_n}{2^{2n}}$ (B) $\frac{1}{{}^{2n}C_n}$ (C) $\frac{1.3.5\dots(2n-1)}{2^n \cdot n!}$ (D) $\frac{3^n}{4^n}$
9. Let $\alpha = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then $B = A + A^2 + A^3 + A^4$, is
- (A) Singular (B) Non-singular
- (C) Skew-symmetric (D) $|B| = 1$
10. A biased coin has $\frac{2}{3}$ probability of landing heads. If the coin is flipped 50 times, then the probability that the number of heads is zero or even is-
- (A) $\frac{3^{50} + 2^{50}}{2 \cdot 3^{50}}$ (B) $\frac{3^{50} + 1}{2 \cdot 3^{50}}$ (C) $\frac{3^{50} - 1}{2 \cdot 3^{50}}$ (D) $\frac{3^{50} - 2^{50}}{2 \cdot 3^{50}}$

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. Given $\theta \in \left(0, \frac{\pi}{2}\right)$ such that $\cos \theta = \frac{1}{3}$, then the value of $\left[\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{\cos k\theta}{5^k} \right]$ is equal to
(where $[.]$ denotes greatest integer function)
2. If $\hat{p} \times 2\hat{q} + 2\hat{r} = 2\hat{q} + \hat{p} \times 2\hat{r}$, then $|3\hat{p} + 2\hat{q} - 2\hat{r}|$ is equal to (where $\hat{p}, \hat{q}, \hat{r}$ are unit vectors)
3. If volume of parallelopiped formed by planes $2x + 2y + z = 4$, $2x + 2y + z = 1$, $2x - y + z = 3$, $2x - y + z = 6$, $3x + 4y + 12z = -1$, $3x + 4y + 12z = 12$ is V, then $\frac{7V}{13}$ is

Space for Rough Work

4. Let $F(z) = \frac{z+i}{z-i}$ for all complex numbers $z \neq i$ and consider $z_n = F(z_{n-1})$ for all positive integers n . Given that $z_0 = 2 + i$, So $|z_{2020}|^2$ is equal to
5. If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ is a vector satisfying $\vec{a} \times \vec{u} = \vec{a} \times \vec{b}$ and $\vec{a} \cdot \vec{u} = 0$ then $2|\vec{u}|^2$ is
6. Let 'F' denotes the set of all onto functions from $A = \{a_1, a_2, a_3, a_4\}$ to $B = \{x, y, z\}$. A function 'f' is chosen at random from 'F'. The probability that $f(a_1) = x$ & $f(a_2) = y$ is 'P', where $36P$ is equal to
-

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 In all other cases

1. Let $z = a + ib$ (where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$) such that $|2z + 3i| = |z|^2$.

Column-I		Column-II	
(A)	Number of such z lying on real axis is	(P)	1
(B)	$ z $ can be	(Q)	2
(C)	If $ z $ is maximum $\Rightarrow z = \alpha + i\beta$, then $(\alpha^3 + \beta^3)$ is equal to (where $\alpha, \beta \in \mathbb{R}$ and $i = \sqrt{-1}$)	(R)	$5/2$
(D)	If $ z $ is minimum $\Rightarrow z = x + iy$, then $(x^2 + 2y^2)$ is equal to (where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$)	(S)	3
		(T)	27

Space for Rough Work

2.

Column-I		Column-II	
(A)	Let $f : A \rightarrow A$ be an one-one function where $A = \{1, 2, 3, 4, 5\}$ formed randomly. The probability that it satisfies $f(f(x)) = x \quad \forall x \in A$ is p , where $\frac{1}{p}$ is equal to	(P)	1
(B)	$\sum_{k=0}^{50} \omega^{k!} = \omega + \lambda$, where λ is (where $\omega (\neq 1)$ is complex cube root of unity)	(Q)	2
(C)	If a, b, c are determined by throw of a single dice thrice, given $a^2 + b^2 + c^2 \leq ab + bc + ca$, the probability that a point (a, b, c) determined in 3 dimensional co-ordinate system lies inside the tetrahedron formed by co-ordinate planes and the plane $x + y + z = 12$ is q , where $\frac{1}{q}$ is equal to	(R)	$\frac{60}{13}$
(D)	Length of portion of curve given by $\arg\left(\frac{z-i}{i}\right) = \pi$, $i = \sqrt{-1}$ intercepted by the curve is $z\bar{z} - z - \bar{z} = 1$	(S)	50
		(T)	47

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work